Model- and Data-Based Approaches to Wind Compensation for Airship Drones

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Abstract

In here, you should briefly describe the thesis content in the primary language.

Kurzfassung

In here, you should briefly describe the thesis content in the secondary language.

Acknowledgements

Contents

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Nomenclature

Abbreviations

COG	Center of Gravity
COV	Center of Volume
CS	Coordinate System
EKF	Extended Kalman Filter
KF	Kalman Filter
UKF	Unscented Kalman Filter

Latin Letters

$\boldsymbol{I}$	Inertia Matrix with respect to the COG

Greek Letters

ω	Angular velocity ($\mathrm{rad\,s^{-1}}$)

Indices

$(\cdot)_B$	Quantity describing the body
${}_B(\cdot)$	Quantity in body-fixed coordinate system
${}_E(\cdot)$	Quantity in Earth coordinate system

1. Introduction

This chapter serves as an overview of existing methods and ...

1.1. Problem setting

1.2. Aim of this work

The Aim of this work is to implement an algorithm for wind estimation and compensation for an unmanned airship. The algorithm should be able to reliable estimate wind speeds in conditions relating to real flight scenarios. Those include: Dynamic wind conditions with Wind gusts of max 30 km/h. Trajectories like forward and sideward flight. Rotation on the spot and forward flight while turning. The compensation should effectively minimize the influence of wind on the aircraft in a direct as possible way.

1.3. State of the art / Related work

[1]

1.4. Airship squid

The airship was designed and built by roboloon and consists of a elastic hull, which is usually filled with Helium. For maneuvering the airships has one stern propeller at the back, four fins and four impellers integrated in the fins totalling 9 control surfaces making the airship an overactuated system.

A full table showing the specifications of the airship can be found in A.1

All further work was carried out and evaluated with respect to the Airship squid.

The airship is controlled by a PID controller ... The airship is equipped with an IMU, GPS and a 1D pitot tube measuring the airspeed in the body-fixes x axis. This pitot tube yields the restriction of only measuring airspeeds abve 1.5m/s, below this threshold no measurement is available.

Allocator

1.5. Simulation Environment / Systematic

All algorithms are first implemented in simulation. This simulation poses a model of the airship as derived in Chapter and an implementation of the PID controller. Additionally uncertainties for the forces ref are considered. After successful validation in the simulation a real world flight test is carried out.

2. Fundamentals

2.1. Coordinate Systems

For this work two coordinate systems are considered. Firstly the body-fixed coordinate system, which axes align with the body axes according to the North-East-Down (NED) convention. The origin of the body-fixed CS coincides with the Center of Gravity of the Airship.

Secondly the earth coordinate system. For simplicity a flat earth model is assumed. In this coordinate system . The origin refers to an arbitrary point on the earths' surface and is chosen to be at the inital position of the airship. The Earth CS is inertial, which means

why flat earth?

what?

add graphics

2.2. Airship Model

The airship is modeled using a 6DOF Newton Euler approach. Therefore we assume the airship to be a rigid body with constant mass and model the translational and rotational dynamics accordingly. We want to express the dynamics in body-coordinates, which adds some complexity to the derivatives, as we need to consider rotations, but comes in handy for the computation of acting forces and moments. Also important for the implementation of the Kalman Filter, as we have measurements of that state.

The translational dynamics in body-coordinates are given by

$$m\,{}_B\mathbf{a}_B = {}_B\mathbf{F}_{\text{ext}} \tag{2.1}$$

where ${}_B\mathbf{a}_B$ describes the acceleration of the airship in body-fixed coordinates and ${}_B\mathbf{F}_{\text{ext}}$ represents the sum of forces acting on the COG of the airship, described in body coordinates.

To relate the acceleration of the airship ${}_B\mathbf{a}_B$ to its velocity ${}_B\mathbf{v}_B$ we need to consider the rotational velocity of the body-fixed CS w.r.t. the earth CS. Therefore we apply Eulers Transport Theorem to express the translational dynamics as follows

$${}_B\mathbf{a}_B = {}_B\left(\frac{d}{dt}\mathbf{v}_B\right) = {}_B\dot{\mathbf{v}}_B + ({}_B\boldsymbol{\omega}_{EB} \times {}_B\mathbf{v}_B) \tag{2.2}$$

whereas ${}_B\boldsymbol{\omega}_{EB}$ describes the rotational velocity of the body-fixed CS w.r.t. the Earth CS. As the Earth CS is inertial this simply reduces to the rotational velocity of the airship along its three axes.

$${}_B\boldsymbol{\omega}_{EB} = {}_B\boldsymbol{\omega}_B = \begin{pmatrix} p \\ q \\ r \end{pmatrix} \tag{2.3}$$

Combining these equations in 2.1 yields the following dynamics for the translational motion of the airship in body-fixed coordinates

$$m\,{}_B\dot{\mathbf{v}}_B = -m\left({}_B\boldsymbol{\omega}_B \times {}_B\mathbf{v}_B\right) + {}_B\mathbf{F}_{\text{ext}} \tag{2.4}$$

The rotational dynamics

Add rotational Dynamics

$$todo \tag{2.5}$$

We can combine the translational dynamics 2.4 and the rotational dynamics 2.5 into one equation

$$M\,V = matrix \tag{2.6}$$

where $M = todo$ and $V = todo$. To complete the model we are left to model the acting Forces and Moments. By doing so we add the desired complexity to our assumed rigid body with constant mass, resulting in a way more complex body like an airship is one. The quality of the airship model depends heavily on the precise modeling of the acting forces. Any unmodeled Forces and Moments will lead to an error in the model. As it is impossible to capture all Forces and Moments we will later in our design have to consider these uncertainties.

The Forces and Moments acting on the airship can be classified into four categories: Gravitational, Buoyancy, Aerodynamical and Propulsion.

$${}_B\mathbf{F}_{\text{ext}} = F_G + F_B + F_{\text{aero}} + F_{\text{prop}} \tag{2.7}$$

Their individual contribution to the total force and moment is derived in the following.

2.2.1. Gravitational and Buoyancy Forces

2.2.2. Aerodynamical Forces

CITE

As stated in The aerodynamical forces can be further divided upon. As the aim is to achieve an as exact as possible modeling of the airship.

Drag

Added Mass

Viscous Effects

Fins

2.2.3. Propulsion Forces

To be able to compute the Propulsion Components we need to incorporate a Motor Model.

2.3. Wind Model

To make use of these dynamics an accurate wind model is desired. Although the Van Karman Model describes the turbulence spectrum more accurately in most cases this work implements the Dryden Turbulence Model due to its easier implementation. As it results in a linear model. Van Karman is not realizable.

2.3.1. Dryden Turbulence Model

Add comparision plot Random walk and Dryden with appropriate parameter values

The Dryden wind model can be understood as a random walk with memory. . Inputs of the Dryden wind model is white noise. State transition functions

Parameters

The Dryden Model is specified by the following parameters, W20, wind direction and L and sigmas. In the 1970s the US Army developed methods to specify parameters for the Wind model. Therefore we can directly relate the L and sigmas to the current flight state according to following equations:

as W20 and wind direction are fully determined by the wind environment these cannot be easily related to the current flight status.

2.4. Kalman Filter

Kalman Filters came up in the 19xxs as a method to ... Before that state estimation methods like observers were existent. The first one was the widely popular Luenberger Observer... Kalman Filters offer the advantage to ... they also work especially well in combination with white noise inputs.

Bayesian Filtering

There exist several methods to extend the KF framework to nonlinear systems. 2 of these are explained further in the following section.

2.4.1. Extended Kalman Filter

2.4.2. Unscented Kalman Filter

2.5. Disturbance Feedforward

Disturbance Feedforward describes a method to compensate the influence of a known disturbance on the system. Linear Case Nonlinear Case

3. model-based wind estimation and compensation

There exist several estimation methods in control theory. Beginning from simple linear models like the Luenberger Observer etc... The idea is to combine the knowledge about the nonlinear airships dynamics, which have been developed in 2.2 and the wind model into use a state estimation method. Therefore the Kalman Filter proves to be the perfect solution optimally combining dynamics and measurements.

why KF

3.1. Model

First a complete model is required to calculate the state transition function from it. Therefore we augment the six state airship model including any states that have known dynamics and are measured (position + attitude) and the wind states.

State space realization of the wind model. The parameters of the dryden wind model depend on the flight states, so we can easily express them depending on the state. Only the computation of W20 and wind direction is not state dependent, so we have to embody estimators for these. a) konstante schätzwerte erwiesen sich hierfür als ausreichend. Der einfluss von abweichenden angenommenen und tatsächlichen parametern erwies sich als klein b) Die Performance der Wind messung hängt stark von den richtigen parameterwerten ab. Daher wurden schätzmethoden eingerichtet um die Parameter zu schätzen.

estimators (wenn notwendig) nennen, Verweis auf ergebnisse

3.1.1. Discretization

exact discretization of the wind dynamics, approx of model dynamics

3.2. Kalman Filter

To get an Kalman Filter running we need a system in the following form

- discrete state transition function
- discrete measurement function
- Process noise
- Measurement noise

We consider the case of additive process noise, which becomes clear later as we derive the equations.

write down eq, with f, h, Q, R

3.2.1. discrete state transition function

3.2.2. discrete measurement function

The measurement function directly follows from the sensor equipment on board of the airship

3.2.3. Process Noise and Measurement Noise

we have $Q = [Q_model, Q_wind]$ uncertainty through unmodeled forces uncertainty coming from wind -> discretization of covariance

measurement noise from sensor specifications Pitot tube only measurements if $u > 1.5m/s$. To account for this we implement a time varying measurement noise, setting $R_{pitottube} \to \infty$ if airspeed in x-direction is < 1.5. To understand why this works lets look at the computation of the kalman gain. R -> infinity simply means that we fully trust our dynamics and basically do not use the measurement.

RUKU vs Euler

3.3. Implementation of UKF

papers

The UKF obeys 3 Parameters, which mainly influence the spread of the sigma points. Parameters are set to beta = 2, kappa = 0 . The optimal value for alpha is determined by an optimization run.

3.4. Implementation of EKF

3.4.1. Analytical vs Numerical Jacobian

3.4.2. Robustify EKF

...

The EKF is not as robust as the UKF due to several reasons . Robustness is a important desired property of the wind estimation algorithm for the specific use case in this thesis, which is to use it for an wind compensation algorithm. Thus to further improve the robustness an Joseph Form computation of the updated covariance is implemented.

3.5. Compensation

Two approaches are implemented and compared to compensate for the effect of wind. A nonlinear Disturbance Feedforward and NDI. The basic idea of either approach is to extend the current PID control with an compensation term as

$$u = u_{PID} + u_{comp} \tag{3.1}$$

to achieve a partial or full compensation of the wind forces or total forces with u_{comp} and a setpoint tracking behaviour via u_{PID} for the remaining dynamics.

To achieve the fastest possible compensation we compute the compensation directly at an Force level. Adding them up with the desired forces from the PID yields the total desired forces. The allocator turns these desired forces into an control command for the individual motors.

3.5.1. Disturbance Feedforward

The Disturbance Feedforward compensates for the wind induced forces and moments. This means u_{comp} compensates for the additional forces and Moments created by the prevailing wind. Looking at the dynamics one can see that wind affects our airship only via the aerodynamical forces, as they depend on the airspeed Va which is

$$V_a = V_b - V_w \tag{3.2}$$

add dynamics, what is Faero and Fprop?

Fprop describes the propulsion force that is achieved by a command u. Faero is the aerodynamical force. In general Faero and Fprop describe nonlinear functions mapping u and Va to Forces.

combine Forces and Moments into one vector and name it

The Disturbance Feedforward task is to find u_{comp} s.t.

$$F_{\text{aero}}(V_A) + F_{\text{prop}}(u + u_c) = F_{\text{aero}}(V_B) + F_{\text{prop}}(u) \tag{3.3}$$

To further understand what this means lets look how this unfolds assuming a linear system.

$$\begin{aligned} F(V_A) + F_{\text{prop}}(u + u_c) &= F(V_B) - F(V_w) + F_{\text{prop}}(u) + F_{\text{prop}}(u_c) && (3.4)\\ &= F(V_B) + F_{\text{prop}}(u), && (3.5)\\ \Rightarrow \quad F_{\text{prop}}(u_c) &= F(V_w) && (3.6) \end{aligned}$$

in this case F(vw) simply describes the additional forces induced by wind. For nonlinear systems its more complicated.

3.5.2. NDI

As second approach a Nonlinear Dynamic Inversion (NDI) controller is implemented. The idea behind the NDI is to fully compensate the Forces acting on the Airship, basically inverting the dynamics.

$$todo \tag{3.7}$$

Tuning of these approaches works differently. For normal PID tuning disturbances were considered also, which makes the PID hard to tune.

For Compensation the PID was tuned on the undisturbed model (no wind), easier tuning. The results of the compensation were also tested for the PID parameters found before. Problem: tuned without wind doesnt represent the truth as the effect of the wind cannot be fully compensated due to the uncertainties in place and the convergence rate of the KF.

For NDi the model was tuned on the unforced model (so total forces were set to zero)

Given perfect measurements and wind estimate the NDI achieves perfect compensation of all acting forces.

4. data-driven wind estimation and compensation

5. Evaluation and results

First the Performance of the Wind estimation algorithm is assessed in simulation. Later the combination of wind estimation and compensation is compared to the flight envelope without compensation. To evaluate the performance of the Estimation and compensation following flight scenarios are considered

- Forward Flight
  - constant velocity
  - acceleration
- Sideway Flight
  - constant velocity
  - acceleration
- Pure rotation areound the yaw axis
- turn while flying forward (Vorgabe von Vorwärtsgeschwindigkeit und Drehgeschwindigkeit)

5.1. Wind estimation

Following the wind estimation performance is analyzed. First of all we need to find a W20 parameter. We choose W20 = 5, which is justified in the section . ref

Result: Good wind estimation....

5.1.1. Sensitivity to wind parameters

The wind parameters W20 and wind direction cannot be measured or computed directly from the states and thus need to be estimated. In the following the sensitivity to the correct parameter is assessed. Therefore multiple tests are carried out.

First a constant parameter value for W20 was specified, creating the actual wind turbulences in the simulation. The wind estimation performance was assessed for varying estimated W20, reaching from W20 = 0 up to W20 = 30 km/h. For each estimated W20 parameter value 27 wind scenarios are run. In each wind scenario the constant wind, as well as the random seed creating the turbulences, were varied resulting in unique wind conditions for each test. For each test the RMSE between the estimated wind and actual wind are computed individually and the average RMSE along all three axis is computed. Then the results are averaged among all 27 wind scenarios. The results are shown in Table 5.1 and lead to the following conclusions.

a) Even for constant wind $W20_{\text{act}} = 0$, an estimate of $W20_{\text{est}} = 0$ leads to bad results. This can be explained when looking at the computation of the Wind Process noise, for $W20_{\text{est}} = 0$, this results in qwind = 0 , thus not allowing the wind estimate to change at all. So the initial guess will be kept forever? proof deviation, due to the nonperfect modelling and uncertainties in place. (For perfect modelling this deviation would be zero) b) matching W20 do not always imply best performance. Reason herefore are errors in modelling and bigger W20 more flexibility to adapt. So why not chosing W20 = 15 for all? Now look at plots showing differences and oscillations.

really? or just for constant wind = 0

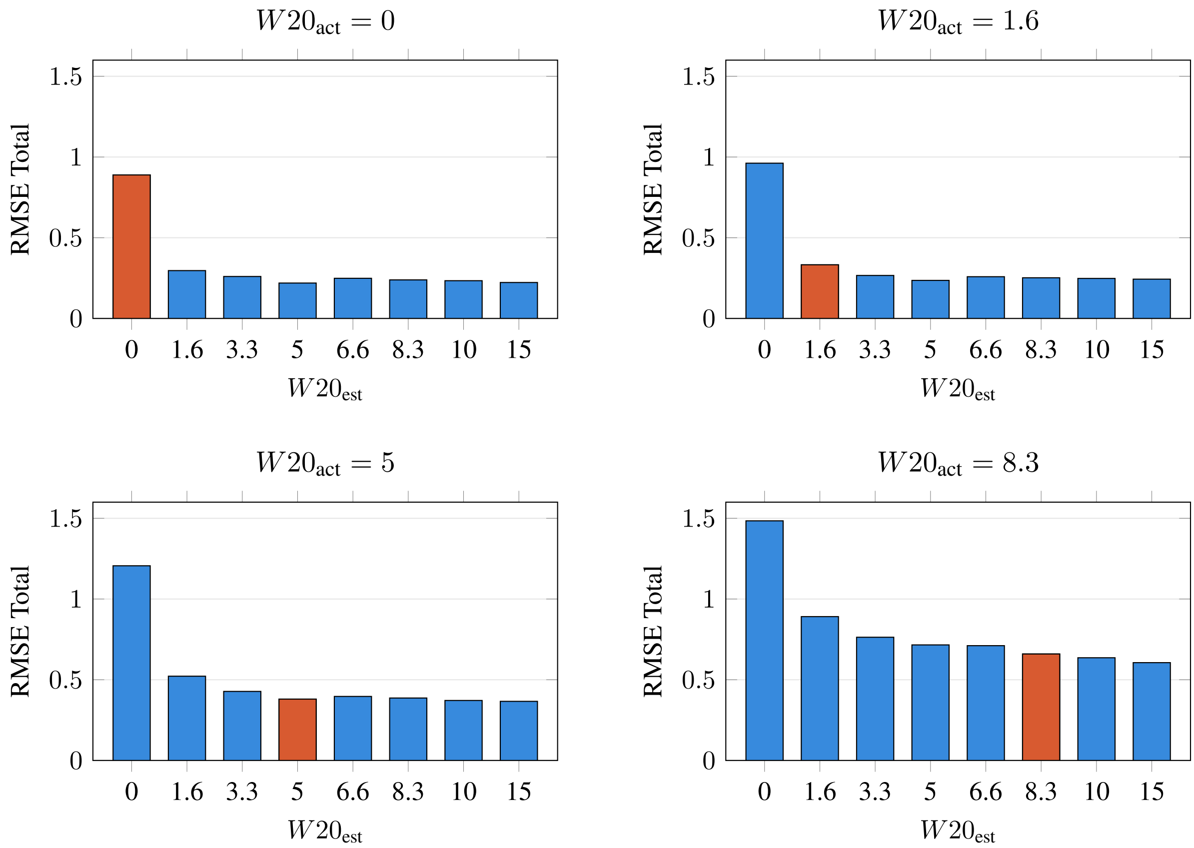

Figure 5.1.: Comparison of Total RMSE Across All Estimated $W20$ Wind Speeds. The orange bar highlights the true configuration scenario where $W20_{\text{est}} = W20_{\text{act}}$.

The results indicate that a despite the real W20 parameter an estimated W20 = 10 yields good results for all real wind conditions created by varying W20. This can be explained as follows: a bigger W20 estimated increases the Process noise, thus giving the wind states more freedom to react to fast changing wind gusts. This comes at a cost which is a more noisy estimation of the wind. W20 estimated = 10 seems to offer a good tradeoff although bigger values may lead to an even lower RMSE. And bigger W20 leads to bigger qwind, this may have bad side-effects.

5.2. Wind compensation

The wind compensation is heavily dependant on the quality of the wind estimation.

5.2.1. Problems with Observability

Due to the lack of sensory equipment to measure airspeed the wind estimation depends heavily on the effect of the aerodynamic forces on the airship to estimate the wind velocities. Because only the aerodynamic Forces are directly dependent on the airspeed and thus on the wind velocity. The Compensation uses the current wind estimate to compute the estimation and thus requires a good estimation of wind. Exactly there lies the problem, if the compensation fully compensates for the effect of wind on the airship, the wind estimation quality automatically decreases, leading to a poor estimation and so on. To break up this cycle we need to guarantee that the information about the effect of the wind is not fully lost. To

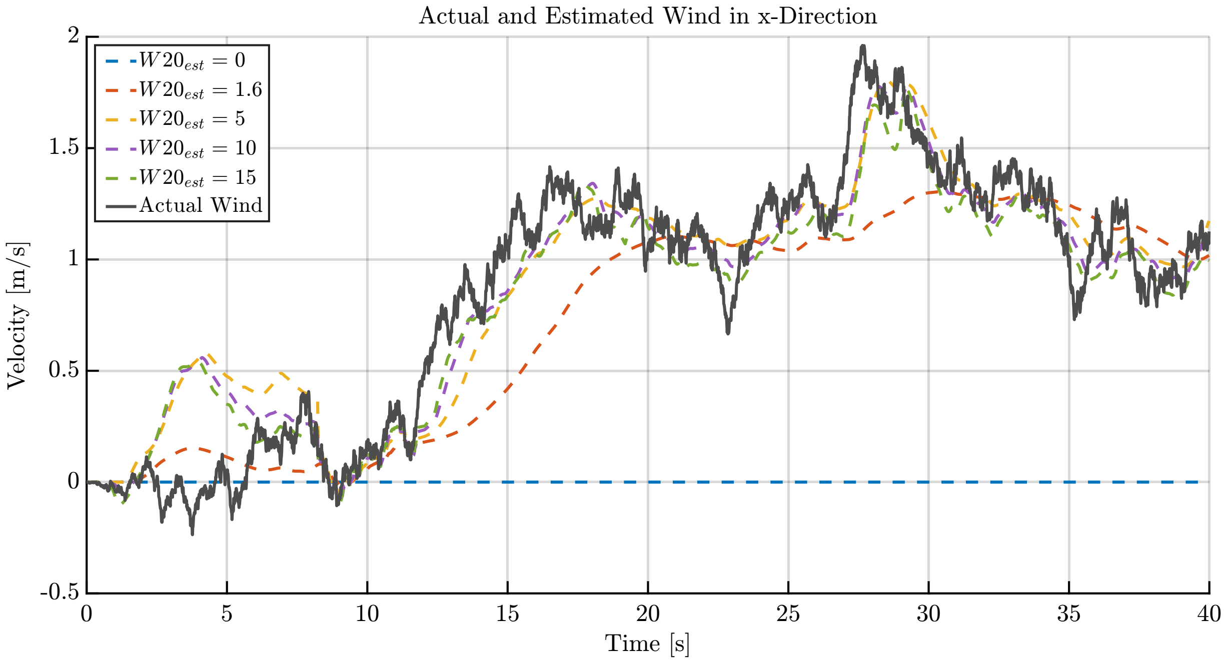

Figure 5.2.: Comparision of wind estimation performance for different estimated W20, W20 actual is 5.

achieve this a gain, manipulating the output of the compensation is introduced. This gain in range 0 to 1 determines how aggressively the wind compensation works and balances the wind compensation with the wind estimation.

5.3. Flight Test

6. Conclusion and Outlook

Outlook

The model-based approach proves to be a powerful tool for wind estimation given a good model of the system is available. The Dryden wind model does not only provide a mathematical description for the wind speeds acting at the center of gravity but also yields a description for the effect of the wind on the angular rates, which are caused by differences in wind magnitudes within the prevailing wind field.

Including the angular rates induced by the differences in wind speeds within the wind field can improve the performance as the airship is especially vulnerable to wind fields considering the airship's large side surfaces. Angular rates of the wind are described by transition functions of order 2 (for the u-direction) and order 3 (for v, w direction) resulting in a increase of one order for every direction compared to wind speeds. It remains to investigate whether the gain in performance can justify the increased complexity.

The Wind estimation proved to be particulary reliable. This information can be used for advanced control and trajectory planning algorithms like MPC to increase the performance compared to the current PID setup.

Bibliography

[1] Walter Fichter and Werner Grimm. *Flugmechanik*. Shaker Verlag, 2009.

A. First Appendix

Write your appendix content here.

A.1. Parameters of the airship squid

Table A.1.: Example Airship Parameters and Specifications

Parameter	**Symbol**	**Value**	**Unit**
Overall length	L	75	m
Maximum diameter	D	18	m
Envelope volume	V	10 000	m^3
Total mass	m	8 500	kg
Payload mass	m_p	1 500	kg
Helium mass	m_{He}	1 700	kg
Cruise speed	v_c	90	km/h
Maximum speed	$v_{\max}$	120	km/h
Maximum altitude	$h_{\max}$	3 000	m
Endurance	t_e	24	h
Propulsion power	P	2×150	kW
Number of engines	N_e	2	–
Buoyant force	F_b	98 000	N
Drag coefficient	C_D	0.045	–
Reference area	A	255	m^2
Moment of inertia (roll)	I_{xx}	1.2×10^5	$kg\,m^2$
Moment of inertia (pitch)	I_{yy}	8.5×10^5	$kg\,m^2$
Moment of inertia (yaw)	I_{zz}	8.9×10^5	$kg\,m^2$
Center of buoyancy offset	z_B	0.8	m

B. Second Appendix

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.